



Transparency Note: Microsoft Discovery

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Table of Contents

What is a Transparency Note?	3
The basics of Microsoft Discovery	3
Introduction	3
Key terms	4
Capabilities	6
System behavior	6
Extensibility	6
Use cases	7
Intended uses	7
Considerations when choosing use cases	8
Limitations	8
Technical limitations, operational factors and ranges	9
System performance	10
Best practices for improving system performance	10
Evaluation of Microsoft Discovery	10
Evaluation methods	10
Evaluation results	11
Evaluating and integrating Microsoft Discovery for your use	11
Learn more about responsible AI	13
Learn more about Microsoft Discovery	13
Contact us	13
About this document	13

What is a Transparency Note?

An AI system includes not only the technology, but also the people who will use it, the people who will be affected by it, and the environment in which it is deployed. Creating a system that is fit for its intended purpose requires an understanding of how the technology works, what its capabilities and limitations are, and how to achieve the best performance. Microsoft's Transparency Notes are intended to help you understand how our AI technology works, the choices system owners can make that influence system performance and behavior, and the importance of thinking about the whole system, including the technology, the people, and the environment. You can use Transparency Notes when developing or deploying your own system, or share them with the people who will use or be affected by your system.

Microsoft's Transparency Notes are part of a broader effort at Microsoft to put our AI Principles into practice. To find out more, see the [Microsoft AI principles](#).

The basics of Microsoft Discovery

Microsoft Discovery is pre-release technology and currently only available to select customers through a private preview. Use of Microsoft Discovery is subject to customer eligibility requirements and acceptance of our Preview Product Terms and User [Code of Conduct](#). It requires a license and is only offered in a limited number of countries. Our goal is to ensure the system remains safe, reliable, and beneficial for all users as we continue to evolve its capabilities.

Introduction

Microsoft Discovery is an AI-powered service that supports the scientific method by helping users source knowledge, generate hypotheses, design experiments, execute them, and analyze results. The system includes the ability to build and orchestrate domain-specific skills and models, along with an autonomous reasoning loop that enhances the entire scientific and development process. Its capabilities can be augmented with customer-specific knowledge and agents.

Microsoft Discovery offers a user experience that is contextually aware and adaptive, driven by natural language. It utilizes large language model (LLM) capabilities to reason over and summarize extensive domain-specific knowledge and customer data. The system can create AI-authored experimentation plans informed by platform capabilities, tools, and models, and enhanced with insights into domain-specific workflows and research trends. It ensures explainability, flexibility, and repeatability, keeping the user in control of the experiment. Additionally, it provides smart execution with adaptive and iterative plans that optimize for time versus cost, or time versus accuracy, based on user guidance. Researchers and developers can manage the process through visibility, controls, and interactivity.

At Microsoft, we are committed to responsible AI. Microsoft Discovery has been developed in line with [Microsoft's AI Principles and Responsible AI Standard](#), in collaboration with responsible AI experts across the company. This document outlines our approach to responsible AI for Microsoft Discovery, detailing the methods used to manage potential risks and misuse, and to secure its benefits for users. As Microsoft Discovery evolves, we continue to learn and improve our responsible AI efforts, updating this document periodically to communicate our evolving processes and methods.

Users are expected to exercise professional judgment when interpreting outputs and making decisions based on system recommendations. Microsoft Discovery is a tool to support—not replace—human expertise.

Key terms

Agentic Framework

An agentic framework refers to the design and utilization of large language models (LLMs) as semi-autonomous agents capable of taking action based on their understanding of natural language input. This framework emphasizes the LLM's ability to perform tasks that resemble decision-making, problem-solving, or goal-oriented behavior, acting in a way that mimics cognitive agency. In Microsoft Discovery, this framework integrates the model with external tools or systems, allowing it to interact with the environment beyond text generation, such as querying databases or controlling software applications.

Classifiers

Machine learning models that sort data into labeled classes or categories. In Microsoft Discovery, classifiers help detect potentially harmful content submitted by users or generated by the system to mitigate misuse or abuse.

Graph RAG

Graph Retrieval-Augmented Generation (Graph RAG) enhances LLMs by integrating graph-based structures for information retrieval and generation. Knowledge is represented as a graph, where nodes represent entities or concepts, and edges represent relationships. The LLM interacts with this graph to retrieve relevant, structured information, improving the relevance and specificity of generated outputs.

Grounding

For certain conversations, Copilot is grounded in web search results, centering its response on high-ranking content from the web and providing hyperlinked citations. Note that user prompts in voice mode will not trigger a web search.

Large Language Models (LLMs)

LLMs are AI models trained on large amounts of text data to predict words in sequences. They can perform tasks such as text generation, summarization, translation, and classification.

Mitigation

Methods designed to reduce potential risks that may arise from using AI features.

Multi-modal Models (MMMs)

MMMs are AI models trained on different types of data, such as text, images, or audio. They can perform tasks like writing text, describing images, recognizing speech, and finding information across different types of data.

Prompts

Inputs in the form of text, images, and/or audio that a user provides to interact with AI features.

Red Teaming

Techniques used by experts to assess the limitations and vulnerabilities of a system and test the effectiveness of planned mitigations. Red team testing includes testers adopting both benign and adversarial personas to identify potential risks.

Small Language Models (SLMs)

SLMs are AI models trained on smaller, more focused amounts of data compared to LLMs. They can perform tasks such as text generation, summarization, translation, and classification, and are often more resource-efficient for specific applications.

System Message

The system message (sometimes referred to as a "meta prompt") guides the system's behavior, aligning it with Microsoft AI Principles and user expectations. For example, it may include instructions like "do not provide information or create content that could cause physical, emotional, or financial harm".

Bookshelf

The Bookshelf is designed to integrate enterprise-specific knowledge, enabling the Copilot to provide insights and recommendations based on proprietary data. It serves as a repository of curated information that can be queried to enhance contexts, plans, and conversations.

Supercomputer

The Supercomputer component of the Microsoft Discovery platform leverages high-performance computing (HPC) and future quantum computing capabilities to run tools, models, and agents for scientific computations and simulations. It supports the execution of complex workflows and provides the computational power needed for advanced scientific research.

Capabilities

System behavior

With Microsoft Discovery, we've developed an innovative approach to supporting the acceleration of scientific discovery. This approach leverages advanced technologies such as language models from Microsoft and from [Azure OpenAI](#).

We implemented safety techniques for the models underlying the new experience to develop a customized set of capabilities and behaviors that provide an enhanced Copilot experience.

In Microsoft Discovery, users can send prompts in natural language, including references to data. Responses are presented in various formats:

- Chat responses in text form (with traditional links to web content as necessary)
- Links or visual representations of complex data structures or computational plans

Grounded Responses and Citations

When responses are provided in text form and grounded in data from the web or customer data, the output may contain hyperlinked citations. These citations allow users to:

- Access the websites or internal documents and databases used to ground the response
- Learn more about the topic

Safety and Model Workflow

When a user enters a prompt in Microsoft Discovery, the user prompt may be handled by one or multiple agents depending on the workflow and agent configuration. For each agent, the prompt is processed according to the following steps:

- 1. Input content classifiers and filtering**
The prompt, conversation history, and system message are sent through several classifiers to filter out harmful or inappropriate content. This step helps mitigate unsafe prompting attempts.
- 2. Large Language Model (LLM) for reasoning**
If the prompt passes the filter, it is sent to an LLM for reasoning. The LLM will decide whether grounding data from the web or bookshelf is required, and which tool will be used to achieve the goal/intention in user prompt.
- 3. Response Generation with Multiple Agent Collaboration**
Upon receiving a user prompt, multiple AI agents may collaborate based on the user's intention. Different agents perform distinct roles to achieve the desired outcome for the user. Depending on the workflow and agent configuration, the final response is generated by the last agent and delivered to the user as the ultimate response to their prompt.

Extensibility

Microsoft Discovery is designed to be extensible, enabling customers to bring their own models (BYOM), integrate proprietary or third-party data through platforms such as AI Foundry or Azure Machine Learning (AML), and build custom agents tailored to their domain-specific needs.

While this flexibility empowers users to enhance the platform's capabilities, it also introduces important responsibilities. Customers are solely responsible for evaluating the appropriateness, accuracy, and compliance of any third-party technologies or datasets they incorporate—this includes Microsoft technologies not explicitly included in the Microsoft Discovery platform.

Customers must also ensure that any such extensions or integrations adhere to the requirements imposed by the Microsoft Discovery Preview Product Terms and User Code of Conduct, including the prohibition on uses that may cause harm.

Use cases

Intended uses

Microsoft Discovery is a platform designed to accelerate discovery by enabling customers to build and integrate advanced agents. The platform itself does not provide end-to-end scenarios out of the box—its capabilities are realized only when customers develop agents tailored to their specific domains and workflows. The scenarios described below are illustrative and assume that customers have created or integrated such agents to enable the described capabilities

- **Create research reports based on internal and external data and papers:** Microsoft Discovery helps users conduct background research on various domains such as chemistry, life sciences, or engineering. This functionality allows researchers to compile comprehensive reports by sourcing and analyzing relevant data and papers. The list of domains can be extended by building your own agents supporting your specific requirements.
- **Create and run complex computational plans using computational tools and ML models:** Users can design and run experiments combining a wide range of tools. Microsoft Discovery understands how to orchestrate, combine and link such tools together to achieve the user's intent. For instance, a user may want to screen molecule candidates based on specific properties; Copilot will select the best tools, set parameters appropriate to the experiment, and write code and perform the end-to-end computations. Microsoft Discovery optimizes computations for accuracy and cost, supporting efficient and effective experimentation.
- **Interpret computational results:** Microsoft Discovery assists in interpreting results to determine compound efficacy and compare new findings with prior art. It also helps create visualization reports tailored to the specific experiments conducted, providing clear and insightful representations of the data.
- **Compound Risk Profiling and Prioritization:** Microsoft Discovery may support early-stage research by helping users identify and prioritize compounds based on predicted biological interactions and safety-relevant characteristics. Leveraging in-vitro data and domain-specific models, the system surfaces candidates that may warrant closer examination, enabling more informed decision-making in experimental planning.

- **Support chip design workflows:** Microsoft Discovery may support semiconductor design teams by enabling the orchestration of complex design flows across hybrid environments. From architecture specification to Register Transfer Level (RTL) generation, simulation, layout, and back-end validation, the platform may assist in automating toolchains and optimizing job scheduling. When integrated with customer-defined agents and domain-specific tools, Microsoft Discovery could help generate synthesizable RTL from specifications, verify logic, and optimize for power, performance, and area (PPA). The system could also assist in scaling high-throughput simulation workloads and managing compute/storage resources efficiently, while supporting secure collaboration across distributed teams.

Considerations when choosing use cases

We encourage customers to leverage Microsoft Discovery in their innovative solutions or applications. However, here are some considerations when choosing a use case:

- **Prohibited Use and Do No Harm Requirement:** Customers must adhere to the Microsoft Discovery Code of Conduct and Product Terms, which prohibit use of Microsoft Discovery in any manner that supports the design, development, or use of any chemical, biological, radiological, or nuclear weapon; or that could cause harm to individuals, organizations, governments, or the environment. This includes misuse of outputs, unsafe experimentation, or deployment in high-risk contexts without appropriate safeguards.
- **Legal and Regulatory Considerations:** Microsoft Discovery is pre-release technology and does not make any commitments regarding compliance, including compliance with industry-specific regulatory requirements. Customers should evaluate the potential specific regulatory obligations that apply to their industry and use scenarios when using Microsoft Discovery.
- **Data Sensitivity and Access Control:** Microsoft Discovery is pre-release technology and does not make any compliance commitments regarding data handling, including handling of sensitive, proprietary, or regulated data. Customers are responsible for determining the appropriateness of their use in this context.

By keeping these considerations in mind, users can make informed decisions and utilize Microsoft Discovery effectively and responsibly. Please refer any questions to TransparencyNotes@microsoft.com.

Limitations

The limitations of Microsoft Discovery should be considered at the intersection of technology and the human, social, and organizational factors that influence its usage and impact. While Microsoft Discovery offers advanced capabilities for supporting the acceleration of scientific discovery, there are certain limitations to be aware of in deploying it responsibly to minimize potential errors and biases.

Technical limitations, operational factors and ranges

Microsoft Discovery is built on large-scale language models and domain-specific tools designed to support scientific research and experimentation. While the system is optimized for accuracy, explainability, and adaptability, it is still subject to technical limitations that may affect its performance, reliability, or fairness. These limitations may arise from the nature of the training data, the complexity of scientific workflows, or the integration of third-party tools and models. The following are key limitations to consider. These are not mutually exclusive—a single output may reflect more than one limitation or affect different user groups in different ways.

- **Representation bias:** Microsoft Discovery may reflect imbalances in the scientific literature or datasets it draws from. For example, when summarizing research or generating experiment plans, the system might overrepresent dominant perspectives or underrepresent findings. These risks are mitigated through prompt tuning, classifier filters, and transparency in data sources, but users should remain aware of the potential for representational imbalance in outputs.
- **Information reliability:** Microsoft Discovery may generate content that sounds plausible but is factually inaccurate or misrepresents source material. Even when grounded in trusted internal or external data, summaries or interpretations may not fully capture the original context. Users are encouraged to apply domain expertise and verify critical outputs before making decisions based on system responses.
- **Quality of service:** Microsoft Discovery is optimized for English-language scientific content. Performance may vary when interacting with non-English data or less common scientific terminology. Additionally, users relying on audio input may experience reduced accuracy due to factors such as background noise, accents, or transcription errors. These limitations can affect the system's ability to process and respond satisfactorily.
- **Domain-specific knowledge gaps:** Microsoft Discovery is a domain-agnostic platform. When knowledge gaps arise across scientific disciplines, they typically originate from either the limitations of the underlying large language model (LLM) or the scope and quality of the knowledge bases (KBs) that users onboard. In domains with sparse training data or less formalized computational tooling—such as emerging or interdisciplinary fields—these gaps may manifest as reduced accuracy or less relevant outputs.
- **Temporal relevance of data:** Scientific knowledge evolves rapidly. If Microsoft Discovery relies on static or outdated datasets, it may surface obsolete findings or miss recent breakthroughs. This is especially important in fast-moving fields like synthetic biology or AI-driven drug discovery.
- **Toolchain compatibility and orchestration errors:** Microsoft Discovery integrates with various computational tools and models. Mismatches in tool assumptions (e.g., input formats or parameter ranges) can lead to execution failures or misleading results.
- **Customer Data:** Microsoft Discovery is a pre-release technology. Users may inadvertently expose Customer Data when grounding prompts or sharing outputs. Users are responsible for managing data sensitivity and access controls when using the service.

System performance

In many AI systems, performance is often defined in relation to accuracy (i.e., how often the AI system offers a correct prediction or output). With Copilot, we're focused on Copilot as an AI-powered assistant that reflects the user's preferences. Therefore, two different users might look at the same output and have different opinions of how useful or relevant it is to their unique situation and expectations, which means that performance for these systems must be defined more flexibly. We broadly consider performance to mean that the application performs as users expect.

Best practices for improving system performance

To improve system performance and adapt system behavior in Microsoft Discovery, there are several best practices that can be followed. These practices involve adjusting various components and parameters to optimize the trade-offs and meet specific use case requirements. However, it is important to consider the potential impact on different populations to ensure fairness and inclusivity.

- **Interact with the Interface Using Natural, Conversational Language:** Similar to adopting techniques to help people communicate effectively in their day-to-day lives, interacting with Microsoft Discovery as an AI-powered assistant either through text or speech that is familiar to the user may help elicit better results.
- **User Experience and Adoption:** Effective use of Microsoft Discovery requires users to understand its capabilities and limitations. There might be a learning curve, and users should reference various Microsoft Discovery resources (e.g., this document and our FAQs) to effectively interact with and benefit from the service.

Evaluation of Microsoft Discovery

Evaluation methods

Evaluations of Microsoft Discovery were conducted using a combination of synthetic and real-world scientific workflows across domains such as chemistry, materials science, and computational biology. The evaluation environment included Microsoft's internal HPC infrastructure and Azure-based compute clusters, simulating realistic research conditions. Evaluation datasets included both publicly available scientific corpora and proprietary datasets contributed by early preview partners, with appropriate data handling and privacy safeguards in place.

The evaluation focused on validating the system's ability to autonomously generate, execute, and refine scientific hypotheses using domain-specific agents and tools. Red team exercises and scenario-based testing were also conducted to assess robustness, safety, and security.

Evaluation results

Microsoft Discovery demonstrated strong alignment with its intended use cases, particularly in hypothesis generation, experiment planning, and result interpretation. In chemistry and materials science, the system successfully orchestrated multi-step workflows involving molecule screening, simulation, and result summarization. These results demonstrate that the system is fit for purpose in its target domains.

Training and test data were selected to reflect a broad range of scientific inquiry types, including structured and unstructured data, multi-modal inputs, and varying levels of domain complexity. Evaluation outcomes also informed design constraints such as maximum prompt length, grounding requirements, and fallback behaviors for unsupported modalities.

While the evaluation focused on high-priority scientific domains, generalizability to other research areas (e.g., environmental science, neuroscience) is expected but not yet validated. Scenarios involving real-time data ingestion or regulatory-sensitive applications were not tested and are flagged for future evaluation. Benchmarking statistics and model cards will be published as part of the public preview documentation.

Evaluating and integrating Microsoft Discovery for your use

When integrating Microsoft Discovery into your scientific workflows, it is important to evaluate how the system performs in your specific context and to establish appropriate oversight mechanisms. Microsoft Discovery is designed to be flexible and adaptive, but its effectiveness depends on how it is configured, the quality of the data it interacts with, and the expertise of the users guiding it.

When preparing to use or using Microsoft Discovery, we recommend keeping the following best practices in mind:

- **Test with real-world data:** Before deploying Microsoft Discovery in production environments, test it using representative datasets and workflows from your domain. This helps identify edge cases, assess grounding quality, and calibrate expectations for performance.
- **Tune system parameters thoughtfully:** Microsoft Discovery allows users to guide trade-offs between time, cost, and accuracy when generating experimentation plans. Adjusting these parameters can significantly affect outcomes. For example, optimizing for speed may reduce the depth of analysis, while prioritizing accuracy may increase computational cost. Consider these trade-offs in the context of your research goals.
- **Monitor agent behaviour and tool orchestration:** If you are using custom agents or integrating third-party tools, validate that the system is orchestrating them correctly. Misalignments in tool assumptions (e.g., input formats or parameter ranges) can lead to silent failures or misleading results.
- **Ensure human oversight:** Microsoft Discovery is not a replacement for expert judgment. Users should review AI-generated experiment plans, summaries, and interpretations before acting on them. This is especially important in high-stakes scenarios such as compound screening or hypothesis generation, where automation bias could lead to overreliance on system outputs.
- **Support interpretability and reproducibility:** Document the inputs, parameters, and outputs associated with each session. This ensures that results can be reviewed, validated, and reproduced by others in your research team.

- **Training:** Provide training or onboarding materials that explain how Microsoft Discovery works, what it can and cannot do, and how to interact with it effectively. This includes understanding grounding behaviour, prompt design, and how to interpret system responses.
- **Quality measurements:** Evaluate the quality and relevance of outputs using metrics that align with your goals. For example, you may want to assess the reproducibility of experiment plans, the accuracy of literature summaries, or the interpretability of model-generated insights. Consider how trade-offs between speed, cost, and accuracy affect your scenario, and adjust system parameters accordingly.
- **Human Oversight:** Review all content before making decisions or acting based on Microsoft Discovery's responses, as AI can make mistakes. It's important to ensure that the information provided aligns with your specific needs and context.
- **Choice of Language Models:** When building custom agents, carefully evaluate the choice of large language models (LLMs) based on the domain, task complexity, latency, and cost constraints. Microsoft Discovery supports composable architectures, and the effectiveness of an agent may depend on selecting or fine-tuning an appropriate model for the intended scientific or technical context.
- **User-Persona Fit:** Consider the technical proficiency and workflow expectations of the target users (e.g., researchers, computational experts, engineers). Some use cases may require more “human in the loop” while others may benefit from more autonomous behaviors from the AI.
 - Human in the Loop:
 - A materials scientist is using Microsoft Discovery to explore novel alloy compositions. The AI proposes candidate formulations based on thermodynamic models and prior literature. However, the scientist manually reviews each suggestion, adjusts parameters based on lab constraints, and selects which candidates to simulate. Here, the AI system accelerates ideation, but the expert remains in control of validation and decision-making.
 - More Autonomous:
 - A computational engineer sets up a custom agent to run thousands of chip layout simulations overnight. The agent automatically selects simulation parameters, dispatches jobs to the appropriate compute queues, monitors for failures, and filters results based on power and area constraints. The engineer reviews only the top-ranked outputs the next morning. In this case, the AI system operates with minimal human intervention, optimizing for throughput and efficiency.

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